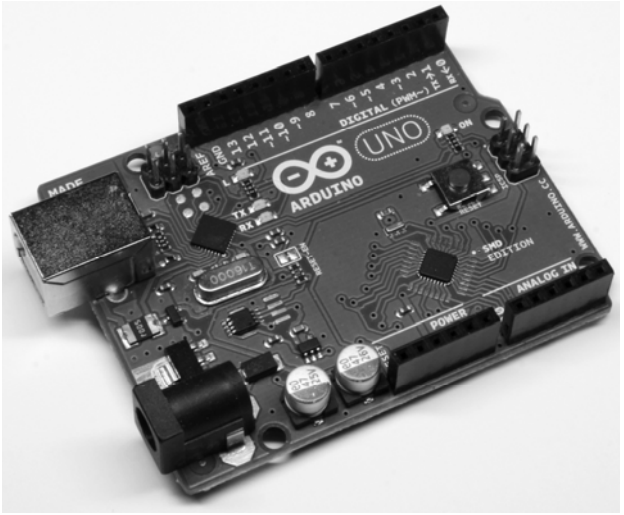


Figure 1-1. *The Arduino Uno interface board, 2011*

This microcontroller comes from a company called Atmel and the chip is known as an AVR. It is slow in modern terms, running at only 16Mhz with an 8-bit core, and has a very limited amount of available memory, with 32 kilobytes of storage and 2 kilobytes of random access memory. The interface board is known for its rather quirky design—just ask the die-hards about standardized pin spacing—but it also epitomizes the minimalist mantra of only making things as complicated as they absolutely need to be. Its design is not entirely new or revolutionary, beginning with a curious merger of two, off-the-shelf reference designs, one for an inexpensive microcontroller and the other for a USB-to-serial converter, with a handful of other useful components all wrapped up in a single board. Its predecessors include the venerable BASIC Stamp, which got its start as early as 1992, as well as the OOPic, Basic ATOM, BASIC-X24, and the PICAXE.

Where all of these precursors are generally closed proprietary products and often require a single computer platform to use, the Arduino development environment is free for all to use and will run on just about any kind of computer that supports Java. The actual hardware board costs a mere USD $30 or EUR €22 and needs nothing more complex than a USB cable to get up and running. This affordable price, nearly half that of its closest competitor, and the board's durable design have led to numerous Arduino-like boards being stitched into embroidery (see Figure 1-2), embedded in pumpkins to be launched through the air by trebuchets, and even sent into outer space in weather balloons.